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Behzad Haki

Education

2020–2024 **PhD in Sound and Music Computing**,
Universitat Pompeu Fabra (UPF), Barcelona, Spain
My dissertation, *Design, Development, and Deployment of Real-Time Drum Accompaniment Systems*, examines the generation of real-time symbolic drum accompaniments with a focus on live improvisation contexts. Over the course of this research, supervised by Dr. Sergi Jordà, I designed, developed, and evaluated three accompaniment systems of increasing complexity, each built around lightweight generative models optimized for real-time performance. Each system was evaluated using a combination of objective measures — including computational performance benchmarks, model quality analyses (generation diversity, interpolation quality, and rhythmic characteristics) — and qualitative evaluations through structured sessions with professional musician Raül Refree, whose iterative feedback directly shaped subsequent system designs, ultimately leading to a series of live public performances with the developed systems. Beyond the primary systems, this work produced several secondary contributions, including NeuralMidiFx, a wrapper for deploying neural networks as VST plugins, two novel datasets (TapTamDrum and El Bongosero), and an exploration of audio-domain adaptations. A hardware Eurorack version of one of the systems was also designed and deployed.

2017–2018 **Master in Sound and Music Computing**,
Universitat Pompeu Fabra (UPF), Barcelona, Spain
My master's studies at the Music Technology Group covered music perception and cognition, real-time interaction, audio programming, digital signal processing, and deep learning. Under the supervision of Sergi Jordà, I completed a thesis on generating basslines interlocked with drum patterns using LSTM networks.

2008–2013 **BASc in Electrical Engineering**,
University of British Columbia (UBC), Vancouver, Canada
I completed a degree in Applied Science with a major in Electrical and Computer Engineering, specializing in Digital Signal Processing (DSP) and acoustics during my final year.

Skills

Programming Software Design, Software Development, Software Testing, Python, C/C++, Matlab, Pure Data, Max/MSP, Concurrent Programming, Multi-Threaded Programming, PyTorch, Libtorch, ONNX, TensorFlow, JUCE, Docker, Linux, VST Plugin Development, Interface Development, Deployment of Neural Networks, Essentia.

Hardware PCB Design, PCB Manufacturing, Circuit Design, Testing, Fabrication, Eurorack Module Development.

Data Science Deep Learning. NLP Techniques for Music Generation. Collection, Curation and Processing of Large-scale Datasets. Data Analysis and Visualization.

Acoustics EASE, CadnaA, ARTA, winMLS, LEAP EnclosureShop and CrossoverShop.

Work Experience

2024–2025 **Research Engineer**, *Universitat Pompeu Fabra*, Barcelona, Spain

• Led research projects in collaboration with undergraduate, master's, and PhD students on interactive sound systems • Designed and developed interactive sonic installations for general public audiences • Translated research outcomes into polished, ready-to-use applications for music production, disseminating research-driven tools to practitioners beyond academia • Curated, maintained, and packaged open-source tools developed at the Music Technology Group for public release • Collaborated with artists on designing and staging live performances integrating interactive generative systems • Designed and conducted experiments for collecting crowd-sourced, open-source datasets from public participants

2014–2019 **Acoustic Engineer/Consultant**, *BAP Acoustics*, Vancouver, Canada

• Conducted noise and vibration measurements • Modeled and simulated noise emissions from existing and proposed future noise sources • Conducted room acoustic measurements • Modeled and simulated acoustics of indoor spaces • Simulated loudspeaker drivers based on electrical and acoustical measurements • Designed loudspeaker cabinets and cross-over circuits for specialized applications • Designed and developed loudspeaker and haptic vibration systems for amusement park ride simulators, creating immersive acoustic experiences emulating real-world vehicles • Simulated outdoor public warning systems • Developed specialized software implementing acoustic standards • Measured noise from railways to monitor the corrugation of tracks • Wrote memorandums and reports

Teaching Experience

Fall 2019 **Introduction To Programming (First Year Engineering)**,
Department of Information and Communications Technologies (DTIC)
Universitat Pompeu Fabra (UPF)
This introductory course provides engineering students with a foundational understanding of programming, with a specific focus on Python. Through hands-on exercises, students are acquainted with the basics of Python programming, equipping them with essential coding skills that serve as a stepping stone for more advanced computational tasks.

Winter 2020 **Electronic Music Production Lab (Fourth Year Engineering)**,
Department of Information and Communications Technologies (DTIC)
Universitat Pompeu Fabra (UPF)
Tailored for senior engineering students, this lab-oriented course delves into the realm of electronic music production. Utilizing the Pure Data environment, participants explore the intricacies of audio digital signal processing (DSP), gaining hands-on experience in crafting sound and understanding the underlying technical processes.

Winter 2021, 2022, 2023 **Computer Organization (First Year Engineering)**,
Department of Information and Communications Technologies (DTIC)
A fundamental course for first-year engineering students, Computer Organization offers a deep dive into the architecture and operation of computers. Key topics include memory management techniques, the nuances of assembly language, and other foundational concepts that underpin the organization and functioning of computing systems.

Winter 2021, 2022, 2023 **Computational Music Creativity (Masters)**,
Department of Information and Communications Technologies (DTIC)
Universitat Pompeu Fabra (UPF)
Situated at the intersection of music and technology, this master's level course delves into the world of computational music creativity. Participants are introduced to the basics of deep generative models, providing insights into how advanced algorithms can be leveraged to foster musical creativity and innovation.

Publications

Esteban Gutiérrez*, Behzad Haki*, Carlos Sing Long, Frederic Font, Xavier Serra, and Rodrigo F. Cádiz. Real-time fractional fourier sound synthesis and processing. *Journal of the Audio Engineering Society*, 2026. Under review.

Błażej Kotowski*, Nicholas Evans*, Behzad Haki, Frederic Font, and Sergi Jordà. Exploring Situated Stabilities of a Rhythm Generation System Through Variational Cross-Examination. In *Proceedings of the 6th Conference on AI Music Creativity (AIMC)*, Brussels, Belgium, September 2025.

Anmol Mishra, Satyajeet Prabhu, Behzad Haki, and Martín Rocamora. Learning Microrhythm in Uruguayan Candombe using Transformers. In *Proceedings of the International Computer Music Conference (ICMC)*, Boston, Massachusetts, June 2025.

Nicholas Evans*, Behzad Haki*, and Sergi Jordà. Repurposing a Rhythm Accompaniment System for Pipe Organ Performance. In *Proceedings of the International Conference on New Interfaces for Musical Expression (NIME)*, pages 116–120, Canberra, Australia, June 2025.

Behzad Haki. *Design, Development, and Deployment of Real-Time Drum Accompaniment Systems*. PhD thesis, Universitat Pompeu Fabra, Barcelona, Spain, December 2024.

Nicholas Evans*, Behzad Haki*, Daniel Gomez, and Sergi Jorda. El Bongosero: A Crowd-sourced Symbolic Dataset of Improvised Hand Percussion Rhythms Paired with Drum Patterns. In *Proceedings of the 24th International Society for Music Information Retrieval Conference*. ISMIR, November 2024.

Anmol Mishra, Behzad Haki, Satyajeet Prabhu, and Martín Rocamora. Groove Transfer VST for Latin American Rhythms. In *Presented at 25th International Society for Music Information Retrieval Conference (ISMIR)*. ISMIR, November 2024.

Nicholas Evans*, Behzad Haki*, and Sergi Jorda. GrooveTransformer: A Generative Drum Sequencer Eurorack Module. In *Proceedings of the International Conference on New Interfaces for Musical Expression (NIME)*. NIME, September 2024.

Behzad Haki, Błażej Kotowski, Cheuk Lee, and Sergi Jorda. TapTamDrum: A Dataset for Dualized Drum Patterns. In *Proceedings of the 24th International Society for Music Information Retrieval Conference*. ISMIR, Nov 23 2023.

Behzad Haki, Julian Lenz, and Sergi Jorda. NeuralMidiFx: A Wrapper Template for Deploying Neural Networks as VST3 Plugins. In *Proceedings of the 4th International Conference on on AI and Musical Creativity*, Sep 1 2023.

Behzad Haki*, Teresa Pelinski*, Marina Nieto, and Sergi Jorda. Completing Audio Drum Loops with Symbolic Drum Suggestions. In *Proceedings of the International Conference on New Interfaces for Musical Expression (NIME)*. NIME, Apr 29 2023.

Behzad Haki*, Marina Nieto*, Teresa Pelinski, and Sergi Jordà. Real-Time Drum Accompaniment Using Transformer Architecture. In *Proceedings of the 3rd International Conference on on AI and Musical Creativity*. AIMC, September 2022.

Thomas Nuttall, Behzad Haki, and Sergi Jorda. Transformer neural networks for automated rhythm generation. In *Proceedings of the International Conference on New Interfaces for Musical Expression*, Shanghai, China, June 2021.

Behzad Haki and Sergi Jorda. A bassline generation system based on sequence-to-sequence learning. In Marcelo Queiroz and Anna Xambó Sedó, editors, *Proceedings of the International Conference on New Interfaces for Musical Expression*, pages 204–209, Porto Alegre, Brazil, June 2019. UFRGS.

Supervisions

At UPF, I had the opportunity to collaborate with, and along with Dr. Sergi Jordà, co-supervise a diverse group of students. Most of these collaborations led to academic publications at peer-reviewed conferences.

2024-25 **3 Master Students, 1 PhD Candidate**, *UPF, DTIC*

2022-23 **3 Master Students, 2 Undergraduate Student**, *UPF, DTIC*

2020-21 **3 Master Students**, *UPF, DTIC*

2019-20 **3 Master Students, 1 Undergraduate Student**, *UPF, DTIC*

Conference Reviewer

2025 **International Society for Music Information Retrieval Conference (ISMIR)**

2025 **International Computer Music Conference (ICMC)**

2025 **New Interfaces for Musical Expression Conference (NIME)**

2025 **International Conference on on AI and Musical Creativity (AIMC)**

2024 **New Interfaces for Musical Expression Conference (NIME)**

2023 **International Conference on on AI and Musical Creativity (AIMC)**

2020 **New Interfaces for Musical Expression Conference (NIME)**

Datasets

2023-24 **TapTamDrum**,
`https://taptamdrum.github.io/`
A novel dataset of repeated dualizations from four experienced drummers. This collection consists of 1116 dualized drum patterns.

2023-24 **El Bongosero**,
`https://elbongosero.github.io/`
A large dataset of concurrent drum and bongo improvisations collected as part of the ongoing AI exhibition held at CCCB. This dataset consists of over 6000 improvisations from over 3000 participants.

Open-Source Projects

2020-2024 **GrooveTansformer: A Suite of Generative Models**,
`https://github.com/behzadhaki/GrooveTransformer`
This suite encompasses a collection of generative transformer models specifically tailored for the generation of symbolic drum patterns. Built on the transformer architecture, these models leverage advanced deep generative techniques to produce rhythmic sequences. The project is open-sourced and accessible to the global community, inviting collaboration and further exploration.

2023 **GrooveTansformer: Eurorack Module**
In 2023, the 'GrooveTransformer' took a groundbreaking leap, evolving from a purely digital realm to the tactile world of Eurorack modules. Recognizing the potential of merging modern machine learning capabilities with the hands-on, modular approach of Eurorack, this initiative aimed to amplify user engagement and control in music generation.

2023 **NeuralMidiFx**,
`https://neuralmidifx.github.io/`
Addressing the deployment challenges of AI music generators in musician-favored platforms like DAWs, I developed NeuralMidiFx. This open-source template streamlines the integration of neural-based music systems as VST3 plugins. With a focus on ease of use, NeuralMidiFx reduces technical hurdles, empowering researchers to easily share their generative tools with a broader audience.

2021-2023 **Groove2Drum VST**,
`https://github.com/behzadhaki/Groove2DrumVST`
An advanced VST plugin deploying the GrooveTransformer modules, Groove2Drum offers musicians immediate generative drum pattern capabilities for both composition and accompaniment, seamlessly integrating AI-driven rhythms into standard music production workflows.

2020-2023 **TapTamDrum Dataset**,
`https://taptamdrum.github.io/`
Inspired by traditional drumming practices, TapTamDrum offers a dual-voice reduction dataset, capturing the essence of complex drum patterns using hands-only interpretations. Collected from four experienced drummers, this resource simplifies multi-voiced sequences for rhythm analysis and generation, providing a unique lens for rhythmic exploration.

2020-22 **MonotonicGrooveTransformer**,
`https://github.com/behzadhaki/MonotonicGrooveTransformer`
Designed for live musical accompaniment, this system utilizes a transformer encoder model trained to generate drum patterns from rhythmic inputs. Despite its training on short samples, strategic design enables it to accompany extended musical sequences seamlessly in real-time.

2020-22 **TransformerGrooveInfilling**,
`https://transformergrooveinfilling.github.io/`
Utilizing Transformer-based models, this tool augments audio drum loops with fitting symbolic events. By converging raw audio with symbolic transcriptions through a spectral approach, a real-time VST plugin is provided, seamlessly integrating its generative strengths into live production.

Workshops

2023 **NeuralMidiFx Workshop at AIMC 2023**, *University of Sussex, UK*
In this workshop, participants were introduced to NeuralMidiFx, a JUCE-based VST3 wrapper designed to simplify the integration of AI-driven symbolic music generative models into plugins. Tailored for those unfamiliar with plugin development, attendees were walked through the entire deployment process, culminating in the creation of a VST3 plugin using a pre-trained generative model.

Artist Collaborations

2022-2024 **Collaboration with Raül Refree**, *Barcelona, Spain*
A participatory design collaboration in which professional musician Raül Refree was involved from the outset as a key design partner. Through iterative evaluation sessions, his requirements and feedback directly shaped the architecture and interaction design of the GrooveTransformer system, culminating in a series of live public performances.

2023 **Collaboration with Desert (Cris Checa and Eloi Caballé)**, *Barcelona, Spain*
Collaborated with Barcelona-based duo Desert on a studio evaluation of the GrooveTransformer system, gathering qualitative insights on controllability and generative behaviour in a production context.

Showcases

June 2025 **+RAIN Film Festival**, *Barcelona, Spain*
My colleague Nicholas Evans, and *nara is neus*, performed at the +RAIN festival in which the system developed for the Palau Güell live show was used in a different context.

Nov 2024 **Palau Güell Live Show**, *Barcelona, Spain*
The second performance of Raül Refree with the last transformer-based generative system. In this live show, Refree plays the organ at the venue, and the generative system generates accompaniments that will be played on a secondary organ. In this live show, we explore whether and how a generative system can be used in a context for which it was not designed.

Feb 2024 **CCCB Live Show**, *Barcelona, Spain*
The first performance of Raül Refree with the GrooveTransformer system. In this live show, Refree did an improvised performance with the system.

2023 **+RAIN Film Festival**, *Barcelona, Spain*,
`https://www.upf.edu/es/web/rainfilmfest`
In this concert, Nicholas Evans, my colleague and co-developer of the GrooveTransformer Eurorack Module, performed with the developed module

2023 **Sonar+D Music Festival**, *Barcelona, Spain*,
`https://sonar.es/es/actividad/project-area-music-and-sound`
During the Sonar+D festival, we showcased the GrooveTransformer Eurorack module to the general public.

2023-2024 **CCCB: El Bongosero**, *Barcelona, Spain*,
`https://elbongosero.github.io/`
A six-month participatory installation at the Centre de Cultura Contemporània de Barcelona (CCCB), in which over 3000 members of the public engaged with a generative sonic system. Participants actively shaped the system's evolution through their interactions, contributing to an open-source crowd-sourced dataset of over 6000 improvisations collected as part of the ongoing exhibition.

Non-Academic Publications

Haki, Behzad, De Santis, Eric. "Good Communication in Restaurants: Acoustic Capacity". BAP Acoustics Online Blog. May, 2015. `https://bapacoustics.com/good-communication-in-restaurants-acoustic-capacity/`

Haki, Behzad. "How Good is Bluetooth at its best?". Serene Audio Online Blog. November, 2015. `http://www.sereneaudio.com/blog/how-good-is-bluetooth-audio-at-its-best`

Languages

English Fluent
Farsi Fluent